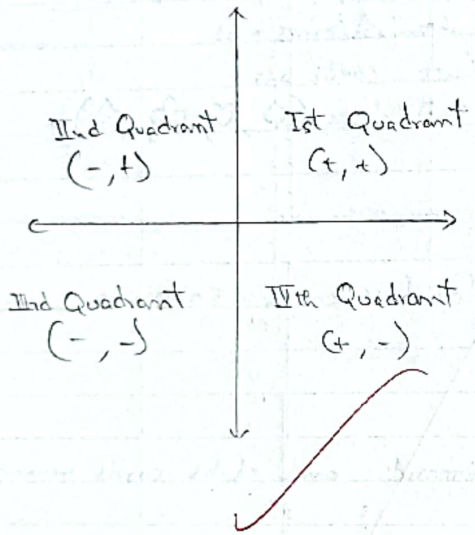

Teacher's Signature: .....

Experiment No. 3

Dated : 7 Aug 2025

Objective : To derive / derivation of Distance Formula

- **Pre-requite Knowledge:**

* **Pythagores Theorem**

Let there be a ΔABC
where 'a' is perpendicular,
'b' is base,
'c' is hypotenuse.

So, $\text{Hypotenuse}^2 = \text{Perpendicular}^2 + \text{Base}^2$

$$c^2 = a^2 + b^2$$

* **Cartession Plan**

*Parallel lines: are those lines that are equidistant from each other and never meet, no matter how much they may be extended in either directions.

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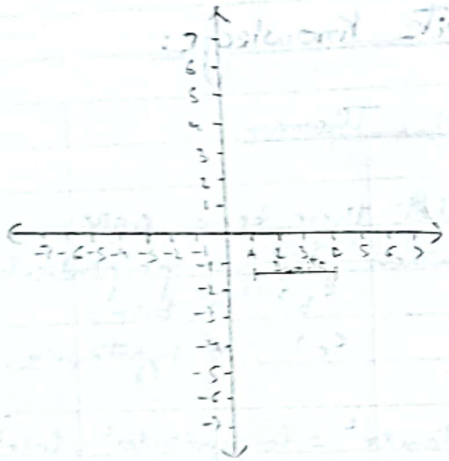
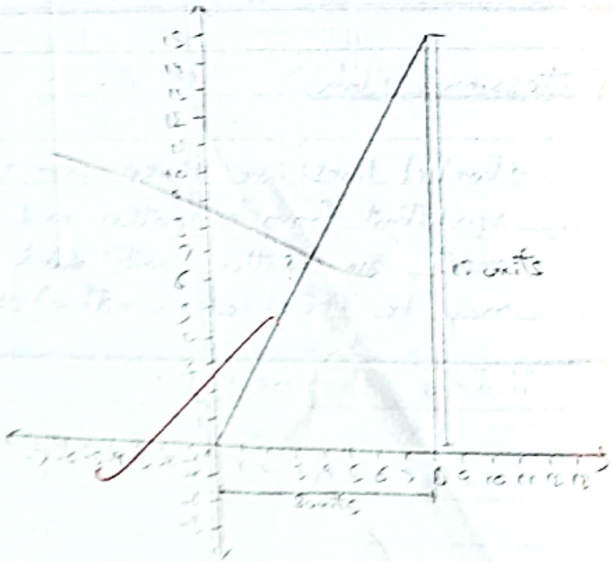
Case 1:



Case 2:



Teacher's Signature: .....

Experiment No. .....

Dated : .....

Objective : .....

- **Material Required:**

  Pencil, scale, pen, eraser, sharpner.

- **Observation and Result:-**

Case 1:

when the points a & b are on x-axis (distance-AB)

$AD = OD - OA$

$= 3$ units

Case 2:

A town is located 36 km east and 15 km north. Find the distance between A & B.

$AB = \sqrt{AD^2 - BD^2}$

$= \sqrt{(8)^2 - (15)^2}$

$= \sqrt{289}$

$= 7$

Hence the distance between the town A and B is 7 units.

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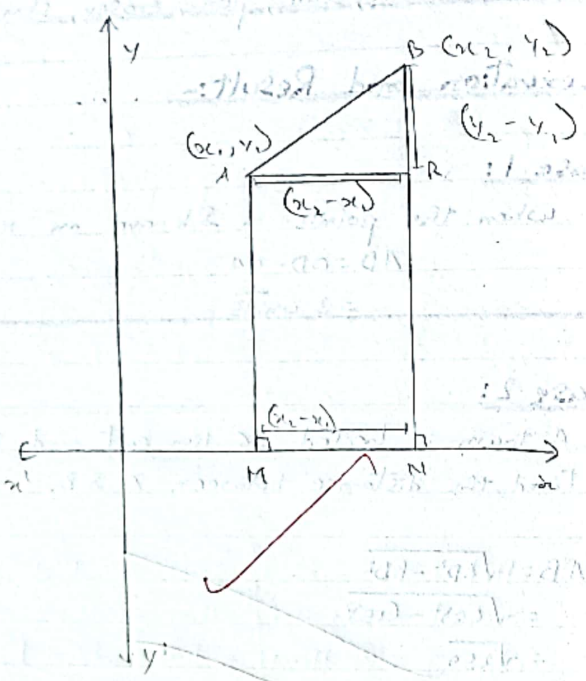
Case 3:



Teacher's Signature: .....

Experiment No. .....

Objective : .....

Dated : .....

Case 3:

When the points are not lying on coordinate axis.

Find the distance between AB
$A(x_1, y_1)$, $B(x_2, y_2)$

Construction:

$AM \perp OX$, $BN \perp OX$
join AR
$\therefore \angle 1 = 90°$ (Concept of Parallel lines)

$\triangle ABR$ is right angled triangle.
By Pythagores theorem
$AB^2 = AR^2 + BR^2$

$AB = \sqrt{AR^2 + BR^2}$
$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Result.
Distance formula is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

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